

SEDAR 88 Gulf of Mexico Red Grouper

Research Recommendations

Recommendations for considerations of future research are provided below and do not indicate any particular order of priority.

Stock Structure

- Better understanding of the population genetics throughout the Gulf of Mexico and connectivity with the Atlantic

Age and Growth

- Investigate methods to better collect age structure samples randomly and systematically from all fishing sectors including surveys - Continue collaboration with ageing facilities throughout the Gulf of Mexico and South Atlantic. These efforts will include the annual reading of references sets for Red Grouper and other reef fish, and annual meetings to review the interpretation of ageing structures and the timing of annual band deposition

Natural Mortality

- Explore more direct approaches to estimating natural mortality (e.g., Mark-recapture approaches (conventional, telemetry, or close-kin))
- Explore ways to better reflect uncertainty around the mortality at age vector

Reproduction

- Continue data collection for maturity, sex transition, and fecundity as detailed in the SEDAR 42 Benchmark Assessment DW Report Recommendations
- Investigate implications of estimating internal growth curve on length and age bases fecundity vector

Discard Mortality

- Continue data collection from observer programs or electronic monitoring programs (e.g., SEDAR88-WP-14)

Commercial Landings

- Explore approaches for assigning uncertainty estimates to commercial landings and revisit estimation of historic landings

CPUE Indices

- Consider developing indices of relative abundance from observer program data (e.g., SEDAR68-AW-04). Observer data would provide finer spatial resolution, a more accurate measure of CPUE, size frequency and discard information

Age and length composition

- Quantify and evaluate appropriate modeling and weighting procedures of length and age compositions to ensure age and length composition inputs are representative of the segment of the population being modeled

Selectivity and catchability

- Further investigate and quantify changes in selectivity/catchability through time to improve fit to the length and age compositions

Surveys

- Improve precision in survey abundance indices by increasing the number of samples - Increase collection of length and age information for compositions
- Investigate effect of survey design changes on overall model time series

Incorporating Red Tide - Explore alternative methods to model red tide mortality with SS, including applying different assumed length- or age-based selectivities based on outputs from ecosystem based models